FAIMS Rectangular Waveform Generator

Operations Manual
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GAA Custom Electronics, LLC
www.GAACustom.com
gaa@owt.com

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WARNING

TO REDUCE THE RISK OF FIRE OR ELECTRICAL SHOCK DO NOT EXPOSE THIS EQUIPMENT TO RAIN OR MOSITURE.

WARNING

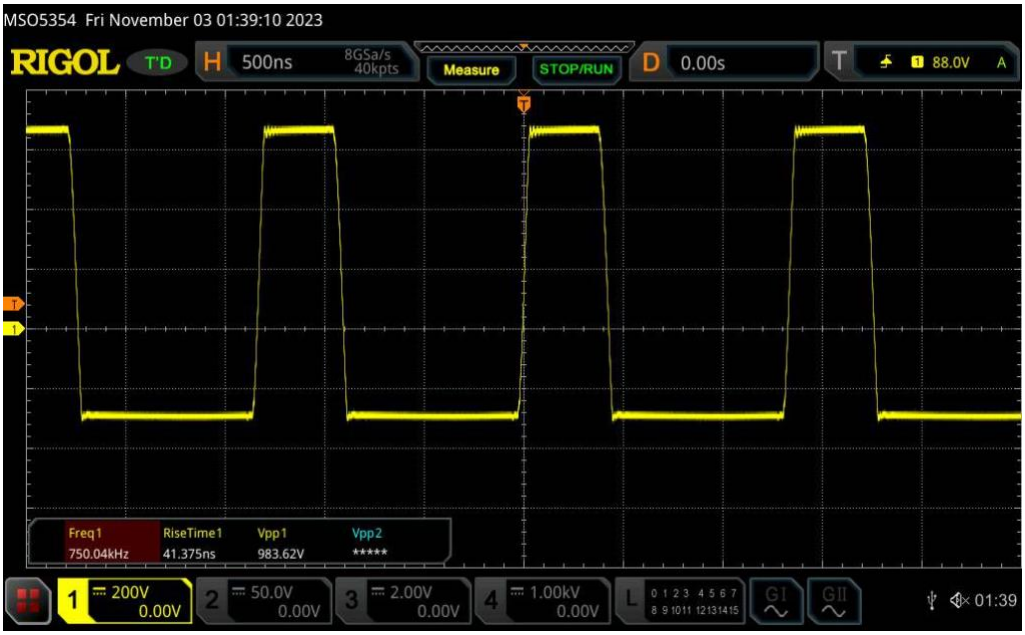
Only qualified electrical workers should operate and install this system.
Avoid spilling liquids onto/into the unit.
Never operate without earth ground.
Do not expose to excessive heat or moisture.
Do not open – there are no user serviceable parts inside.
Do not block the chassis vent slots or the fan inlet.
Do not operate in an explosive environment.

Grounding:

You will find a grounding lug on the rear panel. This lug is used to bond the FAIMS generator chassis to earth ground. It is critical that this is the first connection made and the last connection removed when working with the FAIMS generator.

DO NOT OPERATE THE FAIMS SYSTEM BEFORE YOU READ THIS DOCUMENT!

The FAIMS Rectangular Waveform Generator is designed to generate RF and DC voltages necessary to control a FAIMS device. This system uses high voltage power MOSFETs to generate the RF waveform and DC high voltage amplifiers to generate Bias and CV voltages. A typical output waveform for the FAIMS generator is shown below.



The power MOSFETs were selected to minimize the system power dissipation and as a result of this optimization the turn on and off times are limited to 45 nS as you can see in the scope image above.

The user can define the following waveform parameters:

- Frequency, 100KHz to 1 MHz
- Duty cycle percentage, 5 to 95%
- RF peak to peak voltage, 0 t 1000 or 0 to 1500 depending on model

The power dissipated in the power MOSFETs limits these parameters maximum values. There are two model of the FAIMS generator, 1000V and 1500V. The 1000V model is limited to 1200 watts and the 1500V model is limited to 160 watts. The power will rise linearly with frequency and with the square of the voltage. The duty cycle will not affect the power. With no load the at 750KHz and 1000V the generator will draw 75 watts. This power will increase when the load is attached due to its capacitance.

The power limits are fixed in firmware and if the maximum power is exceeded the system will shut down.

The output signal from the power MOSFETs is capacitively coupled to the output MHV connector though a 3300pF mica capacitor. The average RF output voltage is always zero due to this capacitive coupling. The CV is resistively coupled to the output through 1M ohm of resistance. This allows you to apply DC in the range of -400 to 400 volts. This resistive and capacitive coupling forms an RC filter with a time constant of 20 mS, thus the CV voltage

superimposed on the RF with response to changes with a 20 mS response time. Keep this in mind when selecting scan durations. The same time constant is also present on the Bias output.

The FAIMS generator has internal programmable power supplies to generate DC bias voltage and CV voltage. The CV voltage is superimposed on the FAIMS RF output signal and can be scanned. There are different scan modes described in the Scanning section of this document.

The FAIMS generator can be controlled using the USB interface and the supplied host application as described in the Host interface section of this document. The system can also be controlled via your own application using the command interface described in this document.

Safety Systems

There are a number of safety features designed into the FAIMS generator. Below are a few things to remember:

- Never change any connections when the system is generating a waveform!
- Make sure the FAIMS generator power input is removed when making connections to your FAIMS device.
- Make sure the ground connection is in place at all times.

An Emergency Stop button is located on the rear panel to allow rapid user shut down. The FAIMS generator is equipped with a buzzer to warn the user of the following FAIMS generator events:

- When power is applied to the FAIMS generator the buzzer will beep 3 times.
- When the RF output is enabled, the buzzer will beep 3 times before generating RF and DC voltages.
- If the FAIMS generator exceeds the maximum power limit the status LED with light, the buzzer will beep for 500 mS and the output voltages will be disabled.
- If the FAIMS generator power exceeds the expected power based on user settings and the load capacitance that has been defined then the status LED with light, the buzzer will beep for 500 mS and the output voltages will be disabled.

Scanning

The FAIMS generator supports two CV scan modes:

Toggle

In the toggle mode the CV is toggled between the CV manual setting and the Alt CV value. The toggling can be controlled by defining a duration in mS to remain at each setting or you can provide a Trig input signal to define each CV state. After you and entered the CV parameters you send the toggle command to start the toggle scan.

Linear Ramp

In the linear ramp mode, you define the start and end CV voltages as well as the scan duration in mS. You can start a scan manually using the Scan command or

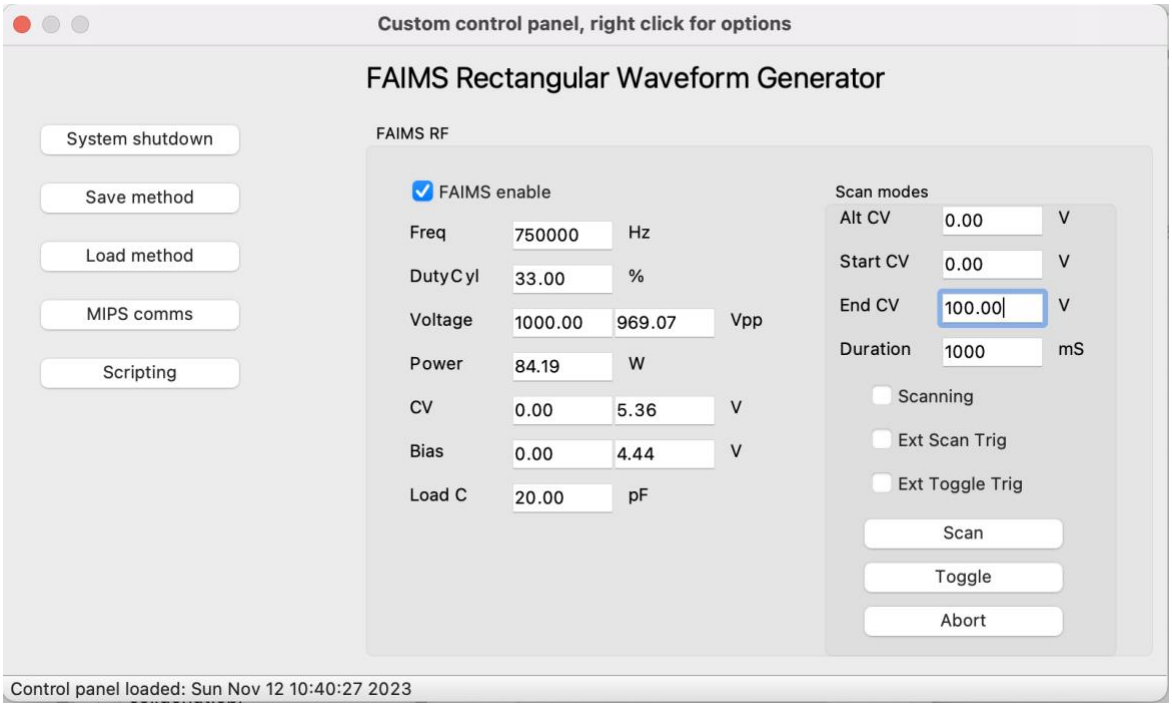
you can trigger a scan to start by applying a trigger pulse to the Trig Input connector. The abort command will allow you to stop a linear scan.

The scan modes can also be controlled using the host interface that is described below.

Host Interface

The MIPS host application and the FAIMSrec.cfg file enables you to control the FAIMS generator using the interface shown below. This is the most user-friendly way to interact with the system and set its parameters. Your system includes a thumb drive that contains the MIPS host application, the FAIMSrec.cfg file and a number of documentation files including this document.

To install the host application just copy the MIPS folder and all its contents to the C drive on your PC, open the MIPS folder and double click on the MIPS application. Make sure the FAIMS generator is powered up and attach the USB cables to your PC, you can then press the “Find MIPS and connect” button, the application will then connect to your system. You can load the control panel using the tools->Select… menu option and select the FAIMSrec.cfg file in your MIPS folder. You can automate this loading process using the properties dialog under the Help menu option. We can help setup the software and show you how to use its capability, just call or email to setup a time.



Front Panel Controls:



24VDC Input

Power is applied through a 4 pin DIN power connector. Pin 1 and 2 are +24VDC and Pin 3 and 4 are return. We have provided a 120-watt 24-volt supply, CUI Inc SDI120-24-U-P51 or equivalent. Please use only the supply provided, the DIN power connectors do not have a consistent pin assignment, different models may plug into the FAIMS generator but could cause damage.

Power

This is a green LED that will light when the 24 volt power supply is connected. Note, the system will always power up with the RF output disabled.

USB

The USB interface is designed to communicate with the MIPS host application. The USB interface appears as a virtual com port on your PC and you should use 9600 or 115,200 baud.

Control

The Control button can but used to enable or disable the RF output. When you press and release this button the system will turn on and off the RF output at the defined RF voltage level.

Status

The Status orange LED will light if there is a system fault. There are two fault conditions, the system exceeds its maximum power limit, or the power exceeds its expected current limit. When these limits exceed the system will also be disabled and you will hear the buzzer beep for 500mS.

RF On

This RF ON red LED is lit with the RF output and DC voltages are enabled.

Pulse In

The FAIMS generator has an internal low voltage waveform generator that drives the high voltage output FETs. An internal jumper can be set to allow you to apply your own low voltage waveform to drive the FETs. This is a 0 to 5 volt input signal. The system is shipped configured to use the internal waveform generator.

Trig Out

The Trig output is a 0 to 5 volt output signal that can be used to synchronize the FAIMS generator with external systems. This output is not functional in the current firmware in the system.

Trig in

The Trig input is a 0 to 5 volt input signal that can be used to trigger or control the CV scanning modes as outlined in the Scanning section of this document.

Buzzer

The buzzer is used to warn the user of changes in the state of the FAIMS waveform generator. You will hear several signals:

- When you apply power to the FAIMS generator the buzzer will beep 3 times.
- When you enable the RF and DC outputs the buzzer will beep 3 times before the voltages are enabled.
- If a fault condition is detected the buzzer will beep once.

Rear Panel Connections:



Emergency Stop

Pressing the Emergency Stop button will remove power from the internal high voltage power supplies and disable the Bias and FAIMS output voltages.

Bias

Bias is a MHV output connector that has the user programmable bias DC voltage. This is a low current -400 to 400 volt signal.

FAIMS

FAIMS is a MHV output connector that has the FAIMS high voltage waveform and the DC CV voltage. This output is designed to drive the capacitive load of the FAIMS device only.

RF on

The RF on indicator is lit when the FAIMS generator is enabled and indicates the system is generating RF and DC voltages on the output connectors.

Ground

You will find a grounding lug on the rear panel. This lug is used to bond the FAIMS generator chassis to earth ground. It is critical that this is the first connection made and the last connection removed when working with the FAIMS generator.

Host Computer Interface:

All commands are ASCII text.
All commands sent to the FAIMS generator are terminated with a carriage return (CR) or line feed (LF) character.
All messages sent from the FAIMS generator are terminated with a CR and LF.
After a command is received (after the CR or LF is received) FAIMS will respond with an ASCII ACK (0x06) then a CR or an ASCII NAK (0x15) then a "?" character and finally a CR.
A NAK indicates the command or its arguments was not understood.
When a NAK is sent the "?" character is sent to inform users communicating with a terminal emulator that the command was not understood.

Many of the commands are described as a pair, a set value command starting with a S and a get value command starting with a G.

Commands:

GVER
: Returns the current version string.

GERR
: Returns an integer error code indicating the reason for the last NAK. This value is never cleared and always indicates the communications last error.

GNAME
: Returns a string that contains the name of the FAIMS generator system.

SNAME,Name
: This command will set the FAIMS generator system name. The Name value is a user defined string. Make sure you issue a Save command to save the changes to flash memory or this name will be lost when power is cycled.

SAVE
: This command will save the FAIMS generator controller parameters to the flash memory.

RESTORE
: This command will read the current settings from the flash memory.

RESET
: This command will cause the FAIMS generator control computer to reboot.

MUTE, Value
: This command will stop all messages from the FAIMS generator system to the host computer. The received commands will still be processed but no

response will be issued. The value parameter turns this mode on and off where:

Value = OFF to turn the responses on and ON to silence the FAIMS generator system.

ECHO,TRUE or FALSE

Setting ECHO true will cause FAIMS generator to include the command with all responses.

GCMDS

This command will cause the FAIMS generator system to display a list of valid commands that it understands.

THREADS

The FAIMS generator system uses a threading control and each module runs on its own Thread as well as a number of threads for control functions. This command displays all the threads and few key parameters. This command is used to monitor the system and check status, mainly a diagnostic tool.

STHRDENA,Thread name,TRUE or FALSE

This command allows you to enable or disable a Thread. This is done by using the thread name and defining it as TRUE or FALSE. If set to FALSE then the thread will stop. This is development and diagnostic tool.

BLOAD

This command will put the FAIMS generator's microcontroller in its boot mode and allow loading new firmware. This is a command used in the factory for upgrading and configuring the system.

The following are waveform setup commands.

SFMENA,TRUE or FALSE

GFMENA

This command will enable the FAIMS generator and resulting in the generation of RF and DC voltage if set to TRUE.

SFREQ,Value

GFREQ

This command will sent the waveform output frequency in Hz, the valid range for Value is 100000 to 1000000.

SDUTY,Value

GDUTY

This command will define the waveform duty cycle in percent, the valid range for this Value is 5 to 95.

SV,Value

GV

This command sets the FAIMS waveform peak to peak voltage level in volts, the valid range is 0 to 1000 or 0 to 1500 depending on the model of your FAIMS generator.

SMAXI,Value

GMAXI

This command allows you to set the maximum output current for each of the two internal supplies that are used to generate the RF signal. This current limit, in mA, is used my each of the supplies and the supplies are in parallel. The total current that can be used in generating the RF signal is twice this value. The valid range for Value is 0 to 50.

SLOADC,Value

GLOADC

This command allows you to define the capacitance of your load in pF. This parameter is used to calculate the expected power based on your other waveform parameters and this capacitance. The system will shut down if the power exceeds this expected value. The valid range for this parameter is 0 to 50.

The following are CV and Bias voltage and scanning commands.

SCV,Value

GCV

This command allows you to set the CV voltage that is superimposed on the FAIMS RF voltage. The valid range for Value is -400 to 400.

GCVV

This command will return the actual CV voltage being generated, in volts, using output voltage monitoring hardware.

SBIAS,Value

GBIAS

This command allows you to set the Bias voltage that you will find on a MHV connector on the rear panel. The valid range for this Value is -400 to 400.

GBIASV

This command will return the actual Bias voltage being generated, in volts, using output voltage monitoring hardware.

SCVALT,Value

GCVALT

This command allows you to define the alternate CV voltage, in volts, that is used by the toggle scan mode. The value range for this Value is -400 to 400.

SALTENA,TRUE or FALSE

GALTENA

This command allows you to enable or disable the toggle scan mode. TRUE will enable the toggle mode.

SCVSTRT,Value

GCVSTRT

This command allows you to define the starting CV voltage for a linear CV scan. The valid range for Value is -400 to 400.

SCVSTP,Value

GSVSTP

This command allows you to define the ending or stopping CV voltage for a linear CV scan. The valid range for Value is -400 to 400.

SDUR,Value

GDUR

This command allows you to define the linear CV scan duration in mS. The valid range for this Value is 10 to 100000.

SSCNTRG,TRUE or FALSE

GSCNTRG

This command allows you to enable the use of the Trig input signal to control toggling or start a scan based on the scan mode you are using.

TOGGLE

This command will start a toggle mode scan.

SCAN

This command will start a linear scan.

STOP

This command will stop any scan in process.

ISSCN

This command will return TRUE if the system is scanning and FALSE if its no scanning.

The following are general monitoring commands.

GHV

This command returns the primary HV power supply voltage.

GHVAUX

This command returns the secondary or auxiliary HV power supply voltage.

GHVI

This command returns the primary HV power supply output current in mA.

GHVIAUX

This command returns the secondary or auxiliary HV power supply output current in mA.

G12V

This command returns the internal 12-volt power supply actual voltage.

GPWR

This command returns the power in watts that is being used by the RF waveform generator power MOSFETs.

Warrantee

GAA Custom Electronics, LLC warrantees the FAIMS system to be free from defects in materials and workmanship and will repair or replace the unit for a period of one year. This warrantee assumes the system is operated in compliance with the procedures and recommendation outlines in this document. GAA Custom Electronics, LLC will also provide

free phone support and firmware bug fixes for up to one year. The addition of new features is not covered in this warrantee.

Liability

The liability of GAA Custom Electronics, LLC hereunder or otherwise is solely and exclusively limited to replacement, repair or credit at the purchase price, as GAA Custom Electronics, LLC may elect, for any product which is returned by Buyer during the applicable warranty period, or services for which timely notice of defect has been given by Buyer, and which are found by GAA Custom Electronics, LLC to be subject to adjustment under this warranty. IN NO EVENT SHALL GAA Custom Electronics, LLC BE LIABLE FOR SPECIAL, INDIRECT, INCIDENTAL OR CONSEQUENTIAL DAMAGES, LOSS OF ANTICIPATED PROFIT OR OTHER ECONOMIC LOSS OR FOR ANY DAMAGES ARISING IN TORT WHETHER BY REASON OF STRICT LIABILITY, NEGLIGENCE OR OTHERWISE.

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OUT:12VDC/2500mA
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EMCO
HV_out
GND
100nMBK6
Control
SIGNAL LAMP